CREDIT DERIVATIVES

Learning Outcome

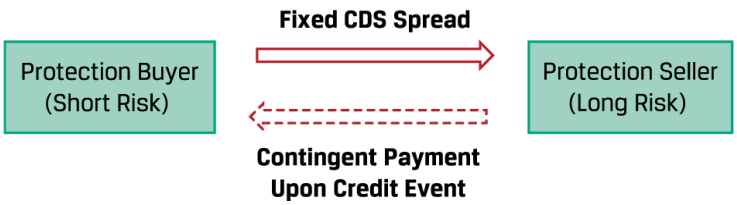
- define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics

Credit derivative contracts are based on a credit underlying, or the default risk of a single debt issuer or a group of debt issuers in an index. The most common credit derivative contract is a credit default swap. CDS contracts allow an investor to manage the risk of loss from issuer default separately from a cash bond. CDS contracts trade based on a credit spread (**CDS credit spread**) similar to that of a cash bond. Credit spreads depend on the probability of default (POD) and the loss given default (LGD), as shown in an earlier lesson. A *higher* credit spread (or higher likelihood of issuer financial distress) corresponds to a *lower* cash bond price, and vice versa.

Despite their name, CDS contracts are contingent claims that share some features of firm commitments. Unlike the call and put options discussed earlier, both the timing of exercise and payment upon exercise under a CDS contract vary depending on the underlying issuer(s). As in the case of a standard interest rate swap, a CDS contract priced at a par spread has a zero net present value, and the notional amount is not exchanged but, rather, serves as a basis for spread and settlement calculations.

In a CDS contract, the credit protection buyer pays the credit protection seller to assume the risk of loss from the default of an underlying third-party issuer. If an issuer **credit event** occurs—usually defined as bankruptcy, failure to pay an obligation, or an involuntary debt restructuring—the credit protection seller pays the credit protection buyer to settle the contract. This contingent payment equals the issuer loss given default for the CDS contract notional amount. Exhibit 10 shows the periodic cash flows under a CDS contract.

Exhibit 10: Periodic Payments under a Credit Default Swap



▶ Expand / collapse Extended Description

The underlying may be a corporate or sovereign issuer, an index of issuers, or a special purpose entity with a portfolio of loans, mortgages, or bonds.

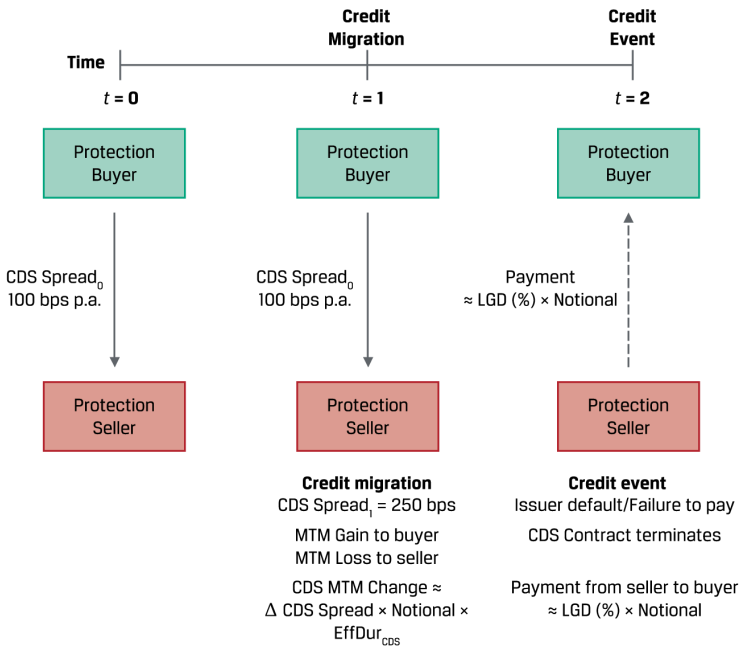
A buyer can use a CDS contract as a hedge of existing credit exposure to the underlying issuer. The credit protection afforded by a CDS is similar to insurance for a buyer with an existing fixed-income exposure to the third-party issuer. The buyer may suffer a loss in value on its fixed-income exposure from the credit event but will receive a payment from the CDS contract that will offset that loss.

A credit protection buyer without the corresponding fixed-income exposure who buys a CDS is seeking to gain from higher credit spreads (which correspond to lower cash bond prices) for an underlying issuer and is therefore short credit risk.

credit event. The contract structure is similar to insurance, with the periodic premium over the life of the contract agreed to upfront and with the timing and size of the loss under the credit event being unknown. The seller's position is therefore similar to that of a long risk position in the issuer's underlying bond.

For example, Exhibit 11 shows the CDS contract for an underlying issuer that experiences credit migration (t = 1) followed by a credit event (t = 2).

Exhibit 11: CDS Contract with Credit Migration and Credit Event



▶ Expand / collapse Extended Description

The protection buyer agrees to pay a fixed spread of 100 bps p.a. at t = 0 for the contract term. As the issuer's CDS spread widens to 250 bps p.a. at t = 1, the buyer gains on the CDS contract due to the low fixed spread paid while the seller loses due to the low fixed spread received relative to the current higher CDS market spread. As for any fixed-income instrument, the effective duration of the remaining contract may be used to approximate the MTM change. An issuer credit event at t = 2 causes the contract to terminate, and the seller must make a payment to the buyer equal to the percentage of loss (LGD) multiplied by the CDS contract notional.

Question Set: Credit Derivatives

1. Determine the correct answer to fill in the blanks: The contingent payment under a credit default swap equals the ____ ____ ____ for the CDS notional amount specified in the contract.

   Hide Solution

   Solution:

   The contingent payment under a credit default swap equals the *loss given default* for the CDS notional amount specified in the contract.

2. Describe how a credit protection seller's position is similar to that of an underlying cash bond investment.

A credit protection seller receives a periodic CDS spread payment in exchange for the contingent risk of payment to the buyer under an issuer credit event. A cash bond investor receives a periodic coupon that incorporates an issuer's credit spread in exchange for a potential loss if the issuer defaults. Under the CDS contract and the cash bond, this potential payment or loss equals the LGD. The credit protection seller's position is therefore similar to that of a long risk position in the issuer's underlying bond.

3. Identify the CDS contract participants that correspond to the following statements:

A. Credit protection buyer	1. Seeks to gain from higher issuer credit spreads
B. Credit protection seller	2. Enters into a derivative contract that transfers the risk of loss from a credit event of an underlying third-party issuer
C. Both a credit protection buyer and a credit protection seller	3. Faces an MTM gain on the CDS contract if the CDS spread of the underlying issuer falls

Hide Solution

Solution:

1. The correct answer is A. A credit protection buyer seeks to gain from higher issuer credit spreads.

2. The correct answer is C. Both the credit protection buyer and credit protection seller enter into a derivative contract that transfers the risk of loss from a credit event of an underlying third-party issuer.

3. The correct answer is B. A credit protection seller faces an MTM gain on the CDS contract if the CDS spread of the underlying issuer falls. The decline in the issuer's CDS spread versus the original fixed spread on the CDS contract means that the protection seller is receiving an above-market spread. This above-market CDS spread more than compensates the seller for the new, lower level of credit risk and results in an MTM gain.

Discuss Filter

Discussion

What do you want to discuss... (required)

BW Exhibit 10: For me, Protection Buyer being Short Risk is unfortunate, because he actually gains from the widening of the credit spread. On the other hand, the protection seller bets that the credit spread will narrow, otherwise the risk of insolvency will decrease - so he should be short risk not the protection buyer. Am I missing something?

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